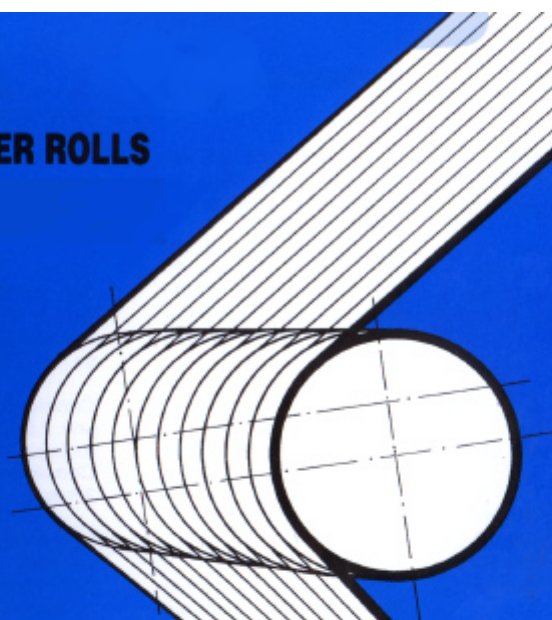


CURVED SPREADER ROLLS



Kickert GmbH • Westring 29 • D-33818 Leopoldshöhe

INSTRUCTIONS FOR OPERATION AND MAINTENANCE

Installation

Optimum effect will be achieved by mounting the spreader roll as shown in Diagram 1 (Fig. 1).

Long entry distance -A-Short output distance -E-The material web should always pass over the concave side of the Bowed Roller.

Dimension 'A'

should al-ways be twice the exit span 'E'

Dimension 'E'

should be approximately two and a half times the diameter of the Bowed Roller.

Note:

When installing the Bowed Roller please ensure that it is aligned both vertically and horizontally onto the material web. As shown in diagram 1 (Fig. 2).

Abb. 1

Fig. 1

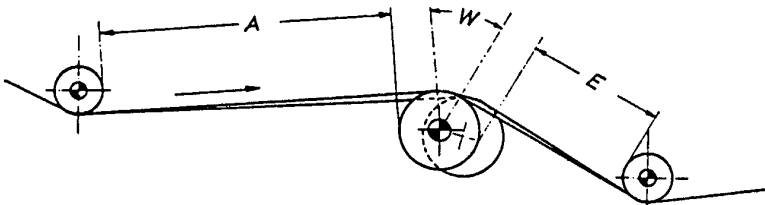
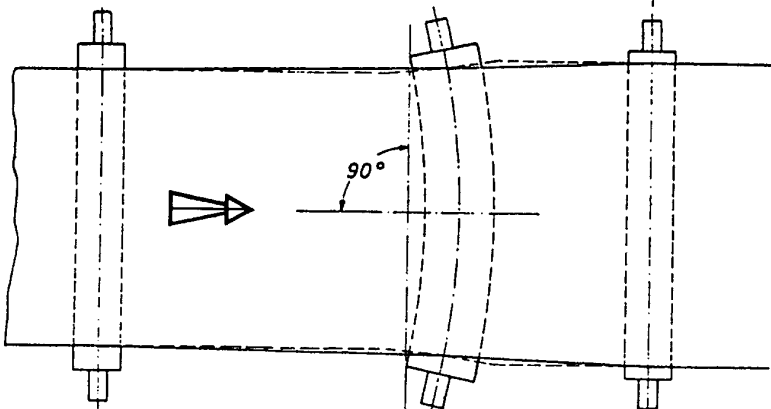


Fig. 2



Wrapping Angle

The wrapping angle -W- is determined by the type of material being processed.

The following angles of wrap are recommended:

Paperindustry

forming fabrics	25-20°
press wires	20-40°
wet felts	30-60°
paper/board	15-30°

Film and Foil Industry

films PP and PA	
1 - 6 Micron	60-90°
8 - 20 Micron	45-60°
more than 20	45°
metalised films	15-20°

Textile Industry

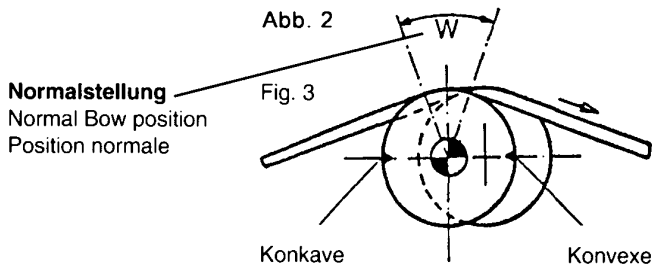
general	45-60°
extensible fabrics	60-90°

Setting

To eliminate creasing, slack edges or slack centre.

Basic setting

To eliminate creasing the angle of entry and exit should be identical. The tension between the centre and edge of the web should be low. Refer to diagram 2 (Fig. 3).



Removal of slack edges

Decrease the angle of wrap around the concave side of the roll until the longitudinal stress of the web is corrected across it's full width.

See (Fig. 4)

Setting of slack centres

Increase the angle of wrap around the convex side of the roll until the longitudinal stress of the web is corrected along it's full width.

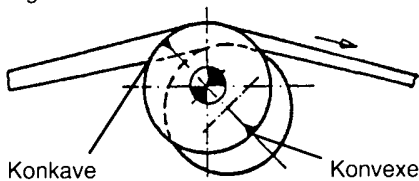
See (Fig. 5)

Stellung bei losen Bahnkanten

Position of slack edges

Position en présence de bords flottants

Fig. 4

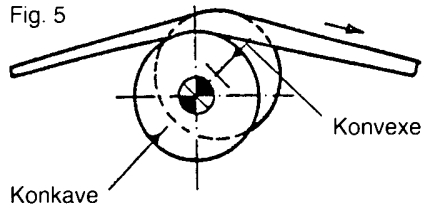


Stellung bei loser Bahnmitte

Position of slack centres

Position en présence de centre de bande moins tendu

Fig. 5

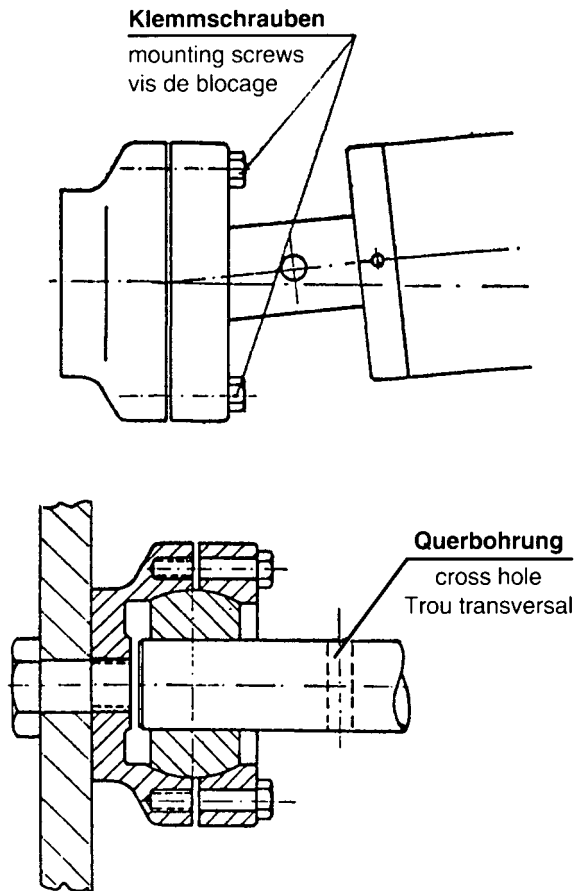


Mounting brackets and adjustment of the roller

Fixed bow roller with simple mounting brackets.

Loosen the mounting screws, adjust the roll by means of a bar in the cross hole. Tighten mounting screws after adjustment.

Abb. 3



Roll with fixed bow and gearbox.

To adjust the roller, loosen the clamping screws on the worm shaft and turn the roller with the hand wheel or ratchet Spanner into its correct position. Tighten the clamping screws again.

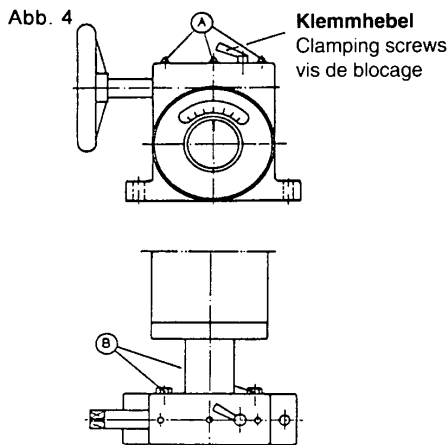
Mounting the machine:

- 1) Loosen the clamping screws **B**.
- 2) Secure the gearbox mounting to the machine frame.
- 3) Tighten the clamping screws on both mounting brackets again.

There is no need for further adjustment.

Grease nipple **A**

Grease monthly with ordinary grease.
(diagram 4)



Roll with Variable curvature adjustable bow

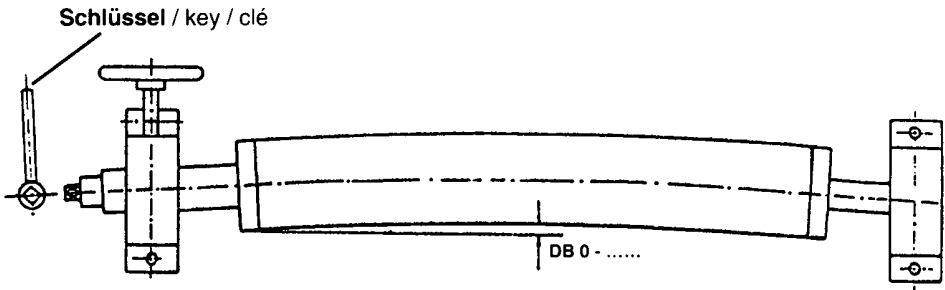
Adjust the roll as described on page 5.

Adjust the amount of bow 'DB' at the front of the square end of the axle using a wrench key or ratchet spanner.

The adjusted bow will now be held.

Record the position on the scale at the front side of the axle.

Abb. 5



Maintenance of the roll

1) Rolls used in a dry working environment are always maintenance free.

2) Rolls in damp working conditions are manufactured with grease labyrinth end seals, and grease nipples.

(diagram 6)

Grease monthly with CALYPSOL L119 or similar.

3) Rolls in wet working conditions are manufactured with an oil chamber system and radial seals. The roller can be greased through the axle. (diagram 7)

Apply grease until it comes out of the labyrinth rings.

Use CALYPSOL U 19 or similar grease.

4) The rollers incorporate sealed for life bearings.

Abb. 6

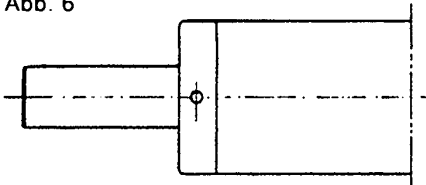
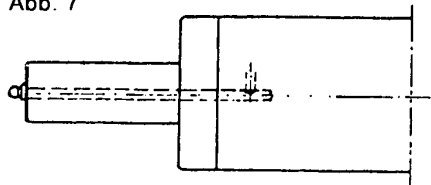


Abb. 7



Roller with hydraulic adjustable curve height

Characteristic features

The essential characteristic features of this type of roller consist of a hand master pump 1, a slave hydraulic cylinder 2, a counter nut 3, a stop screw 4 and a pressure-manometer 5.

Adjustment

Max pressure 100 bar

For setting the curve height the stop screw 4 must be closed first. The rollers axis is curved by pumps with the hand lever.

The ideal bow height for the different webs and applications are determined by watching the web.

The so adjusted curve height is fixed by hand tied screwed of the counter nut 3 (knurled nut marked „Konter“).

Now the constructed pressure can be relieved by opening the stop screw 4. So the system is free of pressure. Thus, any leak losses are eliminated.

To re-adjust the curve height from this position to a higher or lower, the original pressure must be pumped first again, until the counter nut can easily be turned loose.

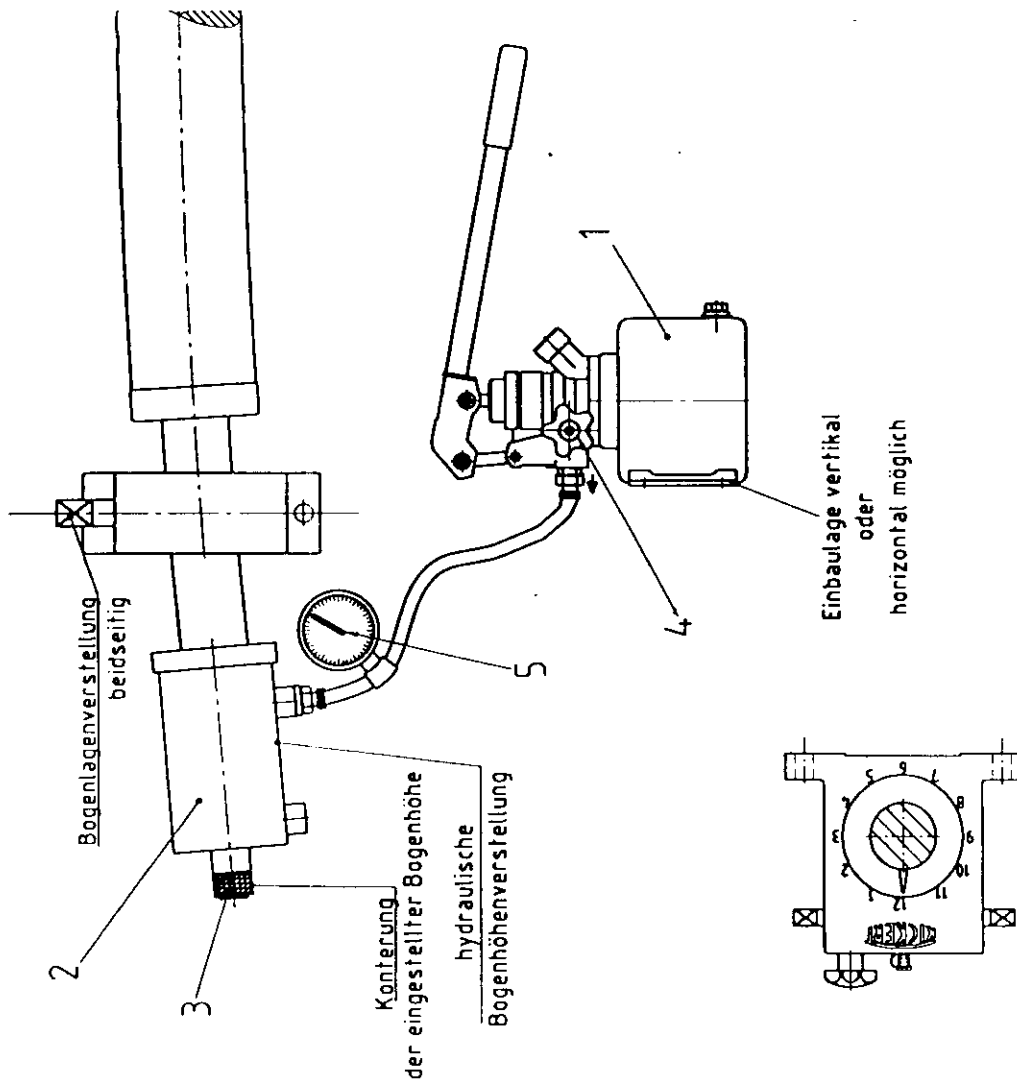
The new curve height can be adjusted and fixed by the same process.

A curve height always corresponds with a corresponding pressure.

On account of the different diameters and lengths of the roller no tables can be named here. These are to be created after your experiences.

Maintenance

The system needs no special maintenance. It is filled with a normal hydraulic oil. The system vents itself.



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Zeich.Nr.

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Angebot./Kom.Nr.

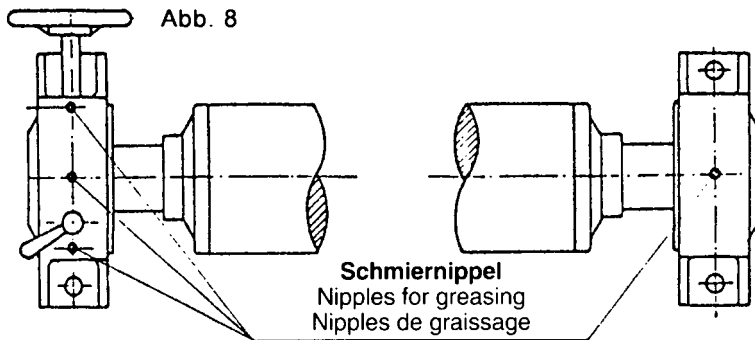
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Maintenance of the swivel worm gear

Greasing is done through the grease nipples (diagram 8).

In dry working conditions once a month is adequate. For wet working conditions once a week is recommended.



Cleaning the roll

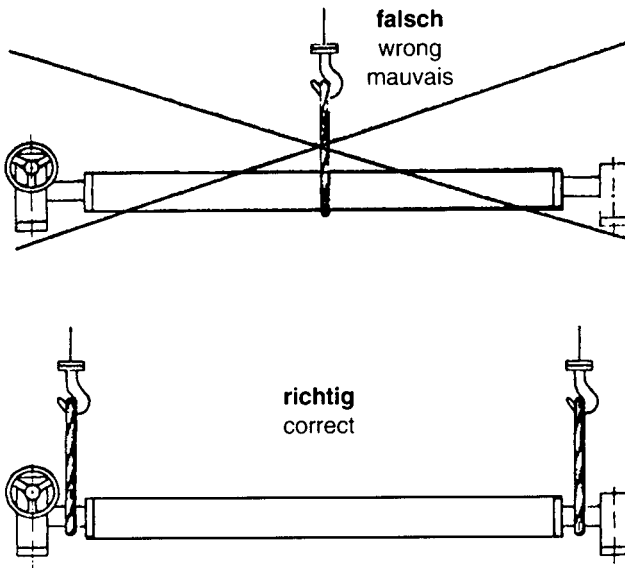
No aggressive solutions or sharp tools may be used because they will damage the rubber sieve.

When cleaning the roll please ensure that water jets are not sprayed directly at the end seals, to prevent the grease being washed out of the labyrinth and seals.

Instructions for the transport

Never lift the roller in centre, as this would lead to damage within the roller.
Always attach the transport belts or slings to the axle ends (diagram 9).

Abb. 9





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